

# HYDROGENATED NBR SYNTHESIS AND PROCESS CONTROL: PATENT LANDSCAPE REVIEW

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## 1. ABSTRACT

This report synthesizes seven patents relevant to hydrogenated nitrile rubber (HNBR) synthesis, with emphasis on how NBR precursors are hydrogenated, how residual unsaturation is controlled, and how molecular weight or processability is tuned. The target material is hydrogenated or partially hydrogenated NBR derived from butadiene/acrylonitrile copolymers, with the desired outcome of improved heat, ozone, and oil resistance while retaining manageable viscosity and low gel content. The patent landscape spans three main process families: conventional solution hydrogenation of NBR using ruthenium catalysts and additives to suppress molecular-weight growth; simultaneous hydrogenation/metathesis routes that reduce molecular weight while hydrogenating in one step; and catalyst-pretreated metathesis/hydrogenation systems that improve hydrogenation rate and preserve polymer molecular weight. A separate application focuses on blending partially hydrogenated NBR with silicone rubber using metathesis catalysts, showing the compatibility of HNBR with reactive blending strategies rather than precursor synthesis itself. Across the hydrogenation patents, the disclosed process variables include hydrogen pressures from 4.137 MPa to 1200 psi, temperatures from 80°C to 145°C, catalyst loadings from 0.015 wt% to 0.25 phr or 500 ppm, and reaction times from 2 to 24 hours depending on catalyst system and target conversion. Additives such as water plus carboxylic acids, or vinyl ethyl ether pretreatment of Ru metathesis catalysts, are repeatedly used to control gelation, molecular-weight increase, or catalyst activity. The metathesis-based patents emphasize Ru/Os catalysts of Grubbs/Hoveyda/Grela/Zhan type, often with co-olefins such as ethylene or vinyl ethyl ether, to achieve lower Mw and narrower PDI. Overall, the evidence supports HNBR synthesis routes that balance high hydrogenation degree with controlled chain scission or minimal crosslinking, and it highlights catalyst/additive engineering as the key lever for scalable HNBR manufacture.

## 2. METHODOLOGY

### PRIMARY PATENT EVIDENCE

#### Patent 1

- Patent Number: US20130303700A1

- Patent Title: Blends of partially hydrogenated nitrile rubber and silicone rubber, ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent does not disclose HNBR synthesis from NBR; it discloses reactive blending of partially hydrogenated nitrile rubber with vinyl-functional silicone rubber in the presence of a metathesis catalyst. The process is carried out in a Brabender mixer with staged addition of catalyst to promote compatibilization and blend morphology control. - Therban® 3467 acrylonitrile content: 34% by weight — Example 1 - Therban® 3467 residual double bond content: 5.5% — Example 1 - Therban® 3467 Mooney viscosity [ML(1 + 4) at 100° C.]: 68 ME — Example 1 - Siloprene® VR vinyl content: 5 mol % — Example 1 - Siloprene® VR number-average molecular weight Mn: 138000 g/mol — Example 1 - Siloprene® VR weight-average molecular weight Mw: 304000 g/mol — Example 1 - Catalyst addition: 0.1 parts by weight — Example 1.2/1.3/1.5/1.7 - Brabender capacity: 350 ml — Example 1 - Fill level: 70% — Example 1 - Rotor speed: 50 min<sup>-1</sup> — Example 1 - Start temperature: 35° C. — Example 1 - Initial kneading time: 2 min — Example 1 - Catalyst addition window: within 5 min — Example 1 - Kneading time: 60 min — Example 1 - Internal temperature: 60° C. — Example 1 - Second/third mixing stage storage: 24 h at room temperature — Example 1 - VMQ content: 10% by weight — SEM selection in Example 1 - VMQ content: 20% by weight — SEM selection in Example 1

**Relevant Experimental Evidence** - Example 1: inventive blends used Grubbs-Hoveyda catalyst added in one or three mixing stages, while comparative blends omitted catalyst. - Example 1: HNBR/VMQ blends were compounded at 35°C then kneaded at 60°C, showing a staged reactive mixing protocol. - Example 1: SEM was used to assess phase morphology of blends containing 10 wt% and 20 wt% VMQ.

**Technical Relevance** This patent is relevant because it uses partially hydrogenated nitrile rubber as the HNBR component and demonstrates how metathesis chemistry can be used in HNBR-containing systems. Although not a precursor hydrogenation patent, it provides process insight into reactive blending and catalyst-assisted compatibility of HNBR with silicone rubber.

## Patent 2

- Patent Number: US10500578B2
- Patent Title: Catalyst compositions and their use for hydrogenation of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019

- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses selective hydrogenation of nitrile rubber in solution using Ru- or Os-based metathesis/hydrogenation catalyst compositions pretreated with a vinyl-containing co-catalyst, especially vinyl ethyl ether. The catalyst is contacted with the co-catalyst before hydrogenation, and in some examples a metathesis step is performed before introducing hydrogen, improving hydrogenation rate and preserving molecular weight. - NBR concentration: 6 wt % — Example 1 - NBR concentration: 3 wt % — Example 2 - Nitrogen bubbling time: 30 minutes — Examples 1-3, 4-9 - Reaction temperature: 120° C. — Examples 1-11 - Hydrogen pressure: 4.137 MPa — Examples 1-7, 9-11 - Agitation speed: 800 rpm — Examples 1-11 - Hydrogenation time: 5 hours — Example 1 - Hydrogenation time: 4 hours — Examples 2, 3, 5, 6, 7 - Hydrogenation time: 3 hours — Examples 8, 9, 10, 11 - Hydrogenation time: 2 hours — Example 4 - Wilkinson's catalyst loading: 15 mg — Example 1 - Triphenylphosphine loading: 18 mg — Example 1 - Catalyst (2) loading: 9 mg — Examples 2, 3, 5, 6, 8, 9, 10, 11 - Catalyst (2) loading: 18 mg — Example 4, 7, 8 - Vinyl ethyl ether pretreatment time: 12 hours — Examples 4, 5 - Vinyl ethyl ether amount: 100  $\mu$ L — Example 5 - Hydrogen pressure: 6.895 MPa — Example 8 - Metathesis time: 1 hour — Example 10 - Metathesis time: 2 hours — Example 11 - VEE amount: 0.5 mL — Example 10 - VEE addition/stirring time: 10 minutes — Example 10 - Catalyst (2) to co-catalyst molar ratio: 42.4 — Table 2 - Catalyst (1) to co-catalyst molar ratio: 72.4 — Table 2 - Hydrogenation degree: 90.3% — Example 1 - Hydrogenation degree: 98.6% — Example 2 - Hydrogenation degree: 92.6% — Example 3 - Hydrogenation degree: 99% — Example 4 - Hydrogenation degree: 95% — Example 5 - Hydrogenation degree: 96% — Example 6 - Hydrogenation degree: 99% — Example 7 - Hydrogenation degree: 98% — Example 8 - Hydrogenation degree: 97% — Example 9 - Hydrogenation degree: 99% — Example 10 - Hydrogenation degree: 99% — Example 11 - Mn: 76286 g/mol — Example 1 - Mw: 260572 g/mol — Example 1 - PDI: 3.42 — Example 1 - Mn: 5560 g/mol — Example 2 - Mw: 14407 g/mol — Example 2 - PDI: 2.59 — Example 2 - Mn: 10103 g/mol — Example 3 - Mw: 19964 g/mol — Example 3 - PDI: 1.98 — Example 3 - Mn: 74495 g/mol — Example 4 - Mw: 229568 g/mol — Example 4 - PDI: 3.08 — Example 4 - Mn: 71220 g/mol — Example 5 - Mw: 224342 g/mol — Example 5 - PDI: 3.15 — Example 5 - Mn: 78125 g/mol — Example 6 - Mw: 256043 g/mol — Example 6 - PDI: 3.28 — Example 6 - Mn: 78514 g/mol — Example 7 - Mw: 222813 g/mol — Example 7 - PDI: 2.84 — Example 7 - Mn: 107058 g/mol — Example 8 - Mw: 261844 g/mol — Example 8 - PDI: 2.44 — Example 8 - Mn: 77473 g/mol — Example 9 - Mw: 219498 g/mol — Example 9 - PDI: 2.83 — Example 9 - Mn: 56391 g/mol — Example 10 - Mw: 140641 g/mol — Example 10 - PDI: 2.49 — Example 10 - Mn: 46540 g/mol — Example 11 - Mw: 105983 g/mol — Example 11 - PDI: 2.28 — Example 11

**Relevant Experimental Evidence** - Example 1: untreated Wilkinson's catalyst gave 90.3% hydrogenation after 5 h with substantial molecular weight retention. - Example 4: VEE-pretreated Catalyst (2) achieved 99% hydrogenation in 2 h, showing strong activity improvement. - Example 5: VEE-pretreated Catalyst (2) gave 95% hydrogenation in 4 h with high Mw retention. - Example 8: higher hydrogen pressure (6.895 MPa) with Catalyst (2) gave 98% hydrogenation in 3 h. -

Example 10: metathesis followed by hydrogenation produced 99% hydrogenation with Mw 140641 g/mol and PDI 2.49.

**Technical Relevance** This is a core HNBR synthesis patent because it directly addresses selective hydrogenation of NBR and demonstrates how catalyst pretreatment and metathesis sequencing can improve hydrogenation efficiency. It is especially relevant for controlling residual unsaturation while maintaining acceptable molecular weight and processability.

### Patent 3

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses simultaneous metathesis and hydrogenation of nitrile rubber to produce low molecular weight HNBR in a single step. The process uses Ru/Os metathesis catalysts with hydrogen and optionally a co-olefin, enabling chain scission and hydrogenation together to lower Mw and Mooney viscosity. - Co-olefin pressure:  $6.9 \times 10^5$  Pa — Examples 1-3 - Cement concentration: 15 wt.-% solid — Examples 1-2 - Cement concentration: 6 wt.-% solid — Examples 3-4 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Co-olefin amount: 150 g 1-hexene — Example 4 / Table 1 - Mn: 85000 g/mol — substrate - Mw: 296000 g/mol — substrate - PDI: 3.50 — substrate - Mn: 73000 g/mol — Experiment 1 - Mw: 189000 g/mol — Experiment 1 - PDI: 2.59 — Experiment 1 - Mooney viscosity: 43 — Experiment 1 - Mn: 60000 g/mol — Experiment 2 - Mw: 136000 g/mol — Experiment 2 - PDI: 2.27 — Experiment 2 - Mooney viscosity: 28 — Experiment 2 - Mn: 31000 g/mol — Experiment 3 - Mw: 59000 g/mol — Experiment 3 - PDI: 1.90 — Experiment 3 - Mooney viscosity: 3 — Experiment 3 - Mn: 55000 g/mol — Experiment 4 - Mw: 111000 g/mol — Experiment 4 - PDI: 2.02 — Experiment 4 - Mooney viscosity: 31 — Experiment 4 - Therban® A3407 Mn: 98000 g/mol — Table 2 - Therban® A3407 Mw: 320000 g/mol — Table 2 - Therban® A3407 PDI: 3.27 — Table 2 - Therban® A3407 Mooney viscosity: 73 — Table 2

**Relevant Experimental Evidence** - Example 1: ethylene co-olefin at 500 psi and 80°C reduced Mw to 189000 g/mol and PDI to 2.59. - Example 2: higher catalyst loading gave Mw 136000 g/mol and Mooney viscosity 28. - Example 3: the lowest Mw and PDI were achieved, with Mw 59000 g/mol and PDI 1.90. - Example 4: 1-hexene co-olefin also enabled strong molecular-weight reduction. - Example 4: the patent reports up to a fivefold reduction in Mw relative to the starting polymer.

**Technical Relevance** This patent is highly relevant because it directly targets low molecular weight HNBR through simultaneous metathesis and hydrogenation. It shows how co-olefin choice, catalyst loading, and temperature can be used to tune Mw, PDI, and Mooney viscosity in HNBR manufacture.

#### Patent 4

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses a hydrogenated carboxylated nitrile rubber with low molecular weight and narrow molecular weight distribution, prepared by olefin metathesis of nitrile rubber followed by hydrogenation. The process is aimed at producing low-Mooney HNBR while preserving the original microstructure better than chemical degradation routes. - MW range: 30000 to 250000 g/mol — Claim 1 - Mooney viscosity range: 3 to 50 — Claim 1 - PDI: less than 2.5 — Claim 1 - Conjugated diene content: 40 to 85 weight percent — Claim 5 - Unsaturated nitrile content: 14 to 50 weight percent — Claim 5 - Unsaturated carboxylic acid content: 1 to 10 weight percent — Claim 5 - Cement concentration: 15 wt.-% solid — Example 1 - Cement concentration: 6 wt.-% solid — Example 3 - Co-olefin pressure:  $6.9 \times 10^5$  Pa — Examples 1-3 - Co-olefin amount: 150 g 1-hexene — Example 4 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 85000 g/mol — substrate - Mw: 296000 g/mol — substrate - PDI: 3.50 — substrate - Mn: 73000 g/mol — Experiment 1 - Mw: 189000 g/mol — Experiment 1 - PDI: 2.59 — Experiment 1 - Mooney viscosity: 43 — Experiment 1 - Mn: 60000 g/mol — Experiment 2 - Mw: 136000 g/mol — Experiment 2 - PDI: 2.27 — Experiment 2 - Mooney viscosity: 28 — Experiment 2 - Mn: 31000 g/mol — Experiment 3 - Mw: 59000 g/mol — Experiment 3 - PDI: 1.90 — Experiment 3 - Mooney viscosity: 3 — Experiment 3 - Mn: 55000 g/mol — Experiment 4 - Mw: 111000 g/mol — Experiment 4 - PDI: 2.02 — Experiment 4 - Mooney viscosity: 31 — Experiment 4

**Relevant Experimental Evidence** - Example 1: ethylene co-olefin at 500 psi and 80°C reduced Mw from 296000 to 189000 g/mol. - Example 2: higher catalyst loading and the same ethylene pressure gave Mw 136000 g/mol and PDI 2.27. - Example 3: the strongest molecular-weight reduction was achieved, with Mw 59000 g/mol and Mooney viscosity 3. - Example 4: 1-hexene co-olefin also produced low-Mw HNBR with Mw 111000 g/mol. - Example 4: the patent demonstrates that metathesis-based degradation can be used to make low-viscosity HNBR.

**Technical Relevance** This patent is relevant because it defines the low-MW HNBR target space and shows a metathesis-based route to reach it. It is especially useful for identifying practical ranges of Mw, Mooney viscosity, and PDI that are compatible with HNBR processing.

## Patent 5

- Patent Number: EP2473280B1
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber using Ru/Os catalysts of Grubbs/Hoveyda/Grela-type structures, optionally with a co-olefin. The process is designed to produce HNBR with lower molecular weight and narrower molecular weight distribution in a single step while maintaining low residual double bond content. - Catalyst amount: 0.057 phr — Example 1 / Table 1 - Catalyst amount: 0.080 phr — Example 1 / Table 1 - Catalyst amount: 0.06 phr — Example 1 / Table 1 - Catalyst amount: 0.055 phr — Example 1 / Table 1 - Metal content: 65.6 ppm — Example 1 / Table 1 - Metal content: 91.3 ppm — Example 1 / Table 1 - Reaction time: 0 min — Table 2 - Reaction time: 60 min — Table 2 - Reaction time: 120 min — Table 2 - Reaction time: 180 min — Table 2 - Reaction time: 240 min — Table 2 - RDB: 100% at 0 min — Table 2 - RDB: 2.0% at 60 min for HNBR 1 — Table 2 - RDB: 0.7% at 60 min for HNBR 2 — Table 2 - RDB: 15.4% at 60 min for HNBR 3 — Table 2 - RDB: 2.9% at 60 min for HNBR 4 — Table 2 - RDB: 0.4% at 120 min for HNBR 1 — Table 2 - RDB: 0.5% at 120 min for HNBR 2 — Table 2 - RDB: 3.1% at 120 min for HNBR 3 — Table 2 - RDB: 0.6% at 120 min for HNBR 4 — Table 2 - RDB: 0.3% at 180 min for HNBR 1 — Table 2 - RDB: 0.3% at 180 min for HNBR 2 — Table 2 - RDB: 1.3% at 180 min for HNBR 3 — Table 2 - RDB: 0.3% at 180 min for HNBR 4 — Table 2 - RDB: 0.6% at 240 min for HNBR 3 — Table 2 - Mw: 191541 g/mol — Table 3 - Mn: 68643 g/mol — Table 3 - PDI: 2.79 — Table 3 - Mooney viscosity: 38.2 — Table 3 - Mw: 135772 g/mol — Table 3 - Mn: 58259 g/mol — Table 3 - PDI: 2.33 — Table 3 - Mooney viscosity: 20.2 — Table 3 - Mw: 195304 g/mol — Table 3 - Mn: 64130 g/mol — Table 3 - PDI: 3.05 — Table 3 - Mooney viscosity: 69.3 — Table 3 - Mw: 144212 g/mol — Table 3 - Mn: 59851 g/mol — Table 3 - PDI: 2.45 — Table 3 - Mooney viscosity: 32.6 — Table 3

**Relevant Experimental Evidence** - Example 1: Grubbs III, Grubbs II and Wilkinson's catalyst were compared at equal metal content. - Example 1: HNBR 1 and HNBR 2 reached very low RDB values by 120 min, while HNBR 3 lagged behind. - Example 1: HNBR 2 gave the lowest Mooney viscosity, 20.2, among the products reported. - Example 1: the patent shows faster hydrogenation for Grubbs III than for Grubbs II or Wilkinson's catalyst. - Example 1: the process yields HNBR with Mw in the 135772-195304 g/mol range and PDI as low as 2.33.

**Technical Relevance** This patent is directly relevant to HNBR synthesis because it combines hydrogenation and metathesis in one process to lower molecular weight while saturating the backbone. It provides comparative catalyst data showing that catalyst identity strongly affects hydrogenation rate, RDB reduction, and final viscosity.

## Patent 6

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses a one-step process for producing low molecular weight HNBR by contacting nitrile rubber with Ru/Os metathesis catalysts in the presence of hydrogen and optionally a co-olefin. The catalyst system is selected to enable simultaneous metathesis and hydrogenation, giving lower Mw and narrower PDI than conventional hydrogenation alone. - Catalyst amount: 0.041 phr — Example 1 / Table 1 - Catalyst amount: 0.045 phr — Example 1 / Table 1 - Catalyst amount: 0.060 phr — Example 1 / Table 1 - Metal content: 65.6 ppm — Example 1 / Table 1 - Co-catalyst amount: 1.0 phr triphenylphosphine — Example 1 / Table 1 - Reaction time: 0 min — Table 2 - Reaction time: 60 min — Table 2 - Reaction time: 120 min — Table 2 - Reaction time: 180 min — Table 2 - Reaction time: 240 min — Table 2 - RDB: 2.5% at 60 min for HNBR 1 — Table 2 - RDB: 7.0% at 60 min for HNBR 2 — Table 2 - RDB: 15.4% at 60 min for HNBR 3 — Table 2 - RDB: 0.6% at 120 min for HNBR 1 — Table 2 - RDB: 4.5% at 120 min for HNBR 2 — Table 2 - RDB: 3.1% at 120 min for HNBR 3 — Table 2 - RDB: 0.5% at 180 min for HNBR 1 — Table 2 - RDB: 3.9% at 180 min for HNBR 2 — Table 2 - RDB: 1.3% at 180 min for HNBR 3 — Table 2 - RDB: 0.4% at 240 min for HNBR 1 — Table 2 - RDB: 3.5% at 240 min for HNBR 2 — Table 2 - RDB: 0.7% at 240 min for HNBR 3 — Table 2 - Mw: 113371 g/mol — Table 3 - Mn: 48352 g/mol — Table 3 - PDI: 2.34 — Table 3 - Mooney viscosity: 33.3 — Table 3 - Mw: 133758 g/mol — Table 3 - Mn: 54595 g/mol — Table 3 - PDI: 2.45 — Table 3 - Mooney viscosity: 41.3 — Table 3 - Mw: 195304 g/mol — Table 3 - Mn: 64130 g/mol — Table 3 - PDI: 3.05 — Table 3 - Mooney viscosity: 69.3 — Table 3

**Relevant Experimental Evidence** - Example 1: Hoveyda and Arlt catalysts outperformed Wilkinson's catalyst in hydrogenation rate. - Example 1: HNBR 1 reached 0.4% RDB at 240 min with Mw 113371 g/mol. - Example 1: HNBR 2 gave Mw 133758 g/mol and Mooney viscosity 41.3. - Example 1: the industrial-norm catalyst produced the highest Mooney viscosity, 69.3. - Example 1: the patent demonstrates that catalyst choice can lower both residual unsaturation and molecular weight.

**Technical Relevance** This patent is relevant because it directly addresses low molecular weight HNBR synthesis via simultaneous hydrogenation/metathesis. It provides a practical catalyst comparison showing that Ru-based catalyst systems can outperform conventional hydrogenation catalysts in both conversion and viscosity control.

## Patent 7

- Patent Number: EP0588099A1
- Patent Title: Catalytic solution hydrogenation of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 1994
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses catalytic solution hydrogenation of nitrile rubber in an organic solvent using divalent ruthenium catalysts, with water and selected organic additives added to suppress molecular-weight increase and gelation. The process is a conventional hydrogenation route rather than a metathesis route, but it is important for controlling HNBR quality and processability. - Autoclave volume: 300 ml — Example 1 - Nitrile rubber acrylonitrile content: about 38 weight per cent — Example 1 - Nitrile rubber butadiene content: about 62 weight per cent — Example 1 - Nitrile rubber Mooney viscosity (ML 1+4 at 100°C): about 50 — Example 1 - Rubber solution concentration: 12 per cent — Example 1 - Catalyst concentration: 500 ppm by weight based on nitrile rubber — Example 1 - Reaction temperature: 145°C — Example 1 - Hydrogen pressure: 1200 psi — Example 1 - Water amount: 1 ml — Example 1 - Water amount: 2 ml — Example 1 / Experiment 10 - Water amount: 0.5 to 2.5 parts by weight per 100 parts by weight of organic solvent — Claims/Description - Hydrogenation temperature range: 80° to 200°C — Description - Preferred hydrogenation temperature range: 120°C to 180°C — Description - Reaction time range: 2 to 24 hours — Description - Preferred reaction time range: 2 to 8 hours — Description - Catalyst concentration range: 0.015 to 2 per cent by weight of nitrile rubber — Description - Preferred catalyst concentration range: 0.015 to 0.15 per cent by weight of nitrile rubber — Description - Experiment 1 reaction time: 4.5 hr — Table 1 - Experiment 1 hydrogenation: 99.7% — Table 1 - Experiment 1 intrinsic viscosity: 2.05 dl/g — Table 1 - Experiment 2 additive concentration: 0.035 g acetic acid — Table 1 - Experiment 2 reaction time: 2.5 hr — Table 1 - Experiment 2 hydrogenation: 99.8% — Table 1 - Experiment 2 intrinsic viscosity: 1.76 dl/g — Table 1 - Experiment 3 additive concentration: 0.040 g propionic acid — Table 1 - Experiment 3 reaction time: 3.5 hr — Table 1 - Experiment 3 hydrogenation: 99.4% — Table 1 - Experiment 3 intrinsic viscosity: 1.78 dl/g — Table 1 - Experiment 4 additive concentration: 0.042 g glycolic acid — Table 1 - Experiment 4 reaction time: 4 hr — Table 1 - Experiment 4 hydrogenation: 99.5% — Table 1 - Experiment 4 intrinsic viscosity: 1.54 dl/g — Table 1 - Experiment 5 additive concentration: 0.052 g chloroacetic acid — Table 1 - Experiment 5 reaction time: 2.5 hr — Table 1 - Experiment 5 hydrogenation: 99% — Table 1 - Experiment 5



intrinsic viscosity: 1.54 dl/g — Table 1 - Experiment 6 additive concentration: 0.089 g adipic acid — Table 1 - Experiment 6 reaction time: 2.5 hr — Table 1 - Experiment 6 hydrogenation: 99.8% — Table 1 - Experiment 6 intrinsic viscosity: 1.61 dl/g — Table 1 - Experiment 7 additive concentration: 0.085 g phenoxyacetic acid — Table 1 - Experiment 7 reaction time: 6.5 hr — Table 1 - Experiment 7 hydrogenation: 99.5% — Table 1 - Experiment 7 intrinsic viscosity: 1.67 dl/g — Table 1 - Experiment 8 additive concentration: 0.043 g glycine — Table 1 - Experiment 8 reaction time: 12 hr — Table 1 - Experiment 8 hydrogenation: 99.7% — Table 1 - Experiment 8 intrinsic viscosity: 1.61 dl/g — Table 1 - Experiment 9 water only: 1 g — Table 1 - Experiment 9 reaction time: 2.5 hr — Table 1 - Experiment 9 hydrogenation: 99.8% — Table 1 - Experiment 9 intrinsic viscosity: 1.87 dl/g — Table 1 - Experiment 10 additive concentration: 0.048 g chloroacetic acid — Table 1 - Experiment 10 reaction time: 2.5 hr — Table 1 - Experiment 10 hydrogenation: 99.7% — Table 1 - Experiment 10 intrinsic viscosity: 1.56 dl/g — Table 1 - Experiment 11 additive concentration: 0.033 g trichloroacetic acid — Table 1 - Experiment 11 reaction time: 2.5 hr — Table 1 - Experiment 11 hydrogenation: 99.7% — Table 1 - Experiment 11 intrinsic viscosity: 1.62 dl/g — Table 1 - Experiment 12 additive concentration: 0.06 g 3-chloropropionic acid — Table 1 - Experiment 12 reaction time: 2.5 hr — Table 1 - Experiment 12 hydrogenation: 99.8% — Table 1 - Experiment 12 intrinsic viscosity: 1.64 dl/g — Table 1 - Experiment 13 additive concentration: 0.08 g 5-chlorovaleric acid — Table 1 - Experiment 13 reaction time: 2.5 hr — Table 1 - Experiment 13 hydrogenation: 99.8% — Table 1 - Experiment 13 intrinsic viscosity: 1.75 dl/g — Table 1 - Catalyst concentration: 20 ppm ruthenium — Example 3 - Rubber amount: 3.25 g — Example 3 - Chlorobenzene amount: 90 ml — Example 3 - THF used as solvent — Example 5

**Relevant Experimental Evidence** - Example 1: water plus selected additives were used with a Ru catalyst to control molecular-weight increase during hydrogenation. - Example 2: multiple organic acids gave high hydrogenation levels around 99-100% with reduced intrinsic viscosity. - Example 3: control experiments without the invention led to crosslinking or very high intrinsic viscosity. - Example 4: alternative Ru catalysts were tested with chloroacetic acid and water. - Example 5: THF solvent also worked, and water plus additive lowered intrinsic viscosity relative to the control.

**Technical Relevance** This patent is relevant because it addresses a major practical issue in HNBR manufacture: molecular-weight increase and gelation during hydrogenation. It provides a process strategy using water and organic additives to preserve processability and control the final HNBR viscosity.

## 3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US10500578B2, EP2473280B1, US9150669B2, and US7772328B2 all use Ru/Os catalyst systems to combine hydrogenation with metathesis or catalyst pretreatment, but US7772328B2 and US7951875B2 emphasize molecular-weight reduction more explicitly than the others.

- EP2473280B1 and US9150669B2 both report low residual double bond contents below about 1% in some examples, with hydrogenation times of 2-4 h and temperatures of 120°C, whereas US10500578B2 reaches 95-99% hydrogenation under similar 120°C / 4.137 MPa conditions after catalyst pretreatment.
- US10500578B2 uses vinyl ethyl ether pretreatment and reports catalyst-to-co-catalyst ratios of 42.4 and 72.4, while EP2473280B1 and US9150669B2 rely on Grubbs/Hoveyda/Arlt/Wilkinson comparisons without the same explicit VEE pretreatment strategy.
- US7772328B2 and US7951875B2 use co-olefins such as ethylene at  $6.9 \times 10^5$  Pa and 1-hexene at 80°C to drive metathesis-based chain scission, achieving Mw as low as 59000 g/mol and Mooney viscosity as low as 3.
- EP0588099A1 is mechanistically different: it is a conventional solution hydrogenation patent that uses water plus organic additives to suppress Mw growth and gelation, rather than metathesis-assisted depolymerization.
- Across the hydrogenation-focused patents, catalyst loadings are low: 0.015-0.15 wt% in EP0588099A1, 0.041-0.060 phr in US9150669B2, 0.055-0.080 phr in EP2473280B1, and 0.057-0.080 phr in US10500578B2.
- The most aggressive viscosity reduction is seen in the metathesis-based low-MW patents (US7772328B2 and US7951875B2), while EP0588099A1 focuses on maintaining viscosity control during hydrogenation rather than intentionally lowering Mw.

## 4. CONCLUSION

The evidence shows two viable synthesis directions for hydrogenated NBR: conventional selective hydrogenation with additives to suppress gelation and molecular-weight growth, and metathesis-assisted hydrogenation to deliberately lower Mw and Mooney viscosity. For high-performance HNBR, the most supported conditions are Ru/Os catalyst systems, 120°C-class temperatures, hydrogen pressures around 4.137 MPa or 1200 psi, and reaction times of 2-4 h when catalyst activity is optimized. For low-MW HNBR, the strongest evidence favors simultaneous hydrogenation/metathesis with co-olefins such as ethylene or 1-hexene, or catalyst pretreatment with vinyl ethyl ether, which can reduce Mw into the 30,000-150,000 g/mol range and PDI toward 2.0-2.5. If the goal is simply to preserve processability during hydrogenation, water plus selected organic acids in EP0588099A1 is the clearest control strategy. Overall, the best-supported synthesis parameters are low Ru loading, controlled temperature near 120°C, hydrogen pressure in the 4-7 MPa range, and catalyst/additive engineering tailored to whether the objective is saturation only or combined saturation plus chain-length reduction.

## 5. REFERENCES

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- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7951875B2/en>
- EP2473280B1 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473280B1/en>
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